

PROJECT REPORT

ON

“FITBUDDY – AI FITNESS PLAN GENERATOR”

Using Gemini Models

A Generative AI Based Fitness Planning Web Application

PROJECT TYPE

Generative Artificial Intelligence / Web Application / Personalized Recommendation System

Submitted in partial fulfilment of the requirements for the award of the degree

BACHELOR OF COMPUTER APPLICATIONS / B.Sc. COMPUTER SCIENCE

GROUP PROJECT

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Department Of COMPUTER APPLICATION

TJS College of Arts and Science

Academic Year 2026

CERTIFICATE

This is to certify that the project report entitled
“FITBUDDY – AI FITNESS PLAN GENERATOR Using Gemini Models”
is a bonafide work carried out by the following students as a Group Project under the guidance and supervision of the undersigned during the academic year 2026.

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Date: 30 September 2026

DECLARATION

We hereby declare that the project entitled “FITBUDDY – AI FITNESS PLAN GENERATOR Using Gemini Models” is an original academic group project carried out by us under the guidance of the faculty member mentioned in this report. The project has been developed for educational purposes and has not been submitted elsewhere for the award of any other degree, diploma, or certificate.

We have made reasonable efforts to understand the technologies used in the project and have documented the requirements, design, implementation, testing, results, and future scope. External technologies and documentation used for the project are acknowledged in the references.

ACKNOWLEDGEMENT

We express our sincere gratitude to the management of TJS College of Arts and Science for providing the facilities and academic environment necessary to complete this project.

We would like to thank the Principal, Head of the Department, project coordinator, and our project guide for their valuable guidance, suggestions, encouragement, and continuous support throughout the development of FITBUDDY.

We also thank our faculty members and friends for their technical assistance, feedback, and encouragement during the planning, development, and testing stages.

Finally, we thank our parents and family members for their continuous support and motivation, which helped us complete this group project successfully.

ABSTRACT

FITBUDDY is a Generative Artificial Intelligence based web application designed to provide personalized fitness-plan suggestions using Gemini models. The system accepts user-provided information such as fitness goals, preferred activity level, available time, workout preferences, and other relevant non-sensitive planning inputs. Based on these inputs, the application generates a structured fitness routine that can include warm-up, workout activities, rest periods, cool-down, and general lifestyle suggestions.

The main objective of FITBUDDY is to demonstrate how Generative AI can be used to create flexible and user-friendly fitness planning assistance. Instead of presenting the same routine to every user, the application uses a prompt-driven approach to adapt the generated plan to the user's stated goals and preferences.

FITBUDDY is designed as an educational technology project rather than a medical or clinical system. The generated information should be treated as general guidance, and users should seek qualified professional advice when appropriate, especially when they have injuries, medical conditions, or other health concerns.

The project combines web development, prompt engineering, Natural Language Processing, recommendation logic, and Gemini API integration. It demonstrates a practical workflow from user input to AI-generated structured output and provides opportunities for future enhancements such as progress tracking, reminders, exercise demonstrations, multilingual support, and integration with wearable devices.

Keywords: Generative AI, Gemini Models, Fitness Planning, Personalized Recommendation, Natural Language Processing, Prompt Engineering, Web Application.

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1. INTRODUCTION

Fitness planning often requires users to organize exercises, repetitions, rest periods, and weekly schedules according to their individual goals and available time. Beginners may find it difficult to create a balanced routine or decide how to structure their exercise sessions. FITBUDDY addresses this planning problem by using Generative AI as an interactive assistant.

The application allows a user to enter a fitness goal such as general fitness, strength development, flexibility, or activity improvement. The user can also provide preferences such as workout duration, frequency, experience level, preferred exercise type, and available equipment. The system processes these inputs and generates a structured plan.

Problem Statement

Many users want a simple starting point for organizing their fitness activities but may not know how to convert their goals and available time into a structured weekly routine. Static plans may not match different preferences. FITBUDDY aims to provide an AI-assisted planning interface that can generate adaptable, clearly formatted suggestions.

Project Idea

FITBUDDY uses a web interface to collect planning information and sends a structured prompt to a Gemini model. The returned content is organized into a weekly or session-based plan. Users can review the result and request modifications such as shorter sessions, different exercise preferences, or a different schedule.

2. PROJECT TYPE AND DOMAIN

Category	Description
Project Type	Generative AI / AI-assisted Web Application
Primary Domain	Artificial Intelligence and Fitness Technology
Sub-Domain	Personalized Recommendation and NLP
AI Model	Gemini family of generative models
Application Area	Fitness planning and activity organization
Interface	Web-based user interface
Target Users	Students, beginners, fitness enthusiasts and general users

Technology Domain

- Generative AI: Creates structured fitness-plan content from user instructions.
- Natural Language Processing: Understands goals, preferences, and planning constraints.
- Prompt Engineering: Converts user inputs into a consistent AI instruction.
- Recommendation Logic: Organizes generated suggestions according to user preferences.
- Web Development: Provides an accessible interface for plan creation and refinement.
- API Integration: Connects the application to the selected Gemini model.

Important Safety Scope

FITBUDDY is an academic AI-planning project and should not be presented as a medical diagnosis, treatment, rehabilitation, or emergency service. AI-generated plans should be reviewed by the user and, when relevant, by a qualified fitness or healthcare professional.



3. EXISTING SYSTEM AND PROPOSED SYSTEM

Existing System

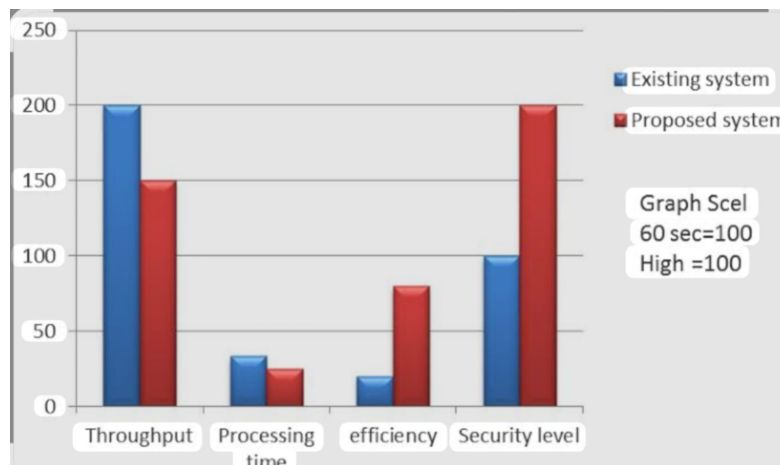
Traditional fitness planning may involve searching for routines online, using static workout applications, manually writing schedules, or consulting a trainer. These approaches can be useful but may require users to compare several resources and manually adapt routines to their available time and preferences.

- Static plans may not match every user's preferences.
- Manual schedule creation can be time-consuming.
- Users may have difficulty adjusting a routine after a change in available time.
- Information may be spread across multiple applications or websites.

Proposed System – FITBUDDY

FITBUDDY uses Gemini models to provide AI-assisted generation of structured fitness-plan suggestions. The user describes their goals and preferences, and the system produces a plan that can be refined interactively.

Existing Approach	FITBUDDY Approach
Static or manually selected routines	AI-assisted personalized planning
Manual schedule modification	Prompt-based refinement
Separate information sources	Single planning interface
Fixed content	Generated according to stated preferences
Manual organization	Structured session/week output



4. OBJECTIVES AND SCOPE

Objectives

1. To develop a user-friendly AI-assisted fitness planning web application.
2. To integrate a Gemini generative model into a practical software project.
3. To generate structured fitness-plan suggestions from user inputs.
4. To organize workouts into clear sessions or weekly schedules.
5. To allow users to refine generated plans through additional prompts.
6. To demonstrate prompt engineering and AI API integration.
7. To provide a foundation for future progress-tracking and fitness features.

Scope

The scope includes collecting user preferences, generating general fitness-plan suggestions, presenting sessions in a structured format, refining generated plans, and optionally saving or exporting plans. The system focuses on general planning and educational demonstration, not clinical or medical advice.

Target Users

- Students learning Generative AI and web application development.
- Beginners looking for a structured starting point for activity planning.
- Users who want to experiment with different schedule formats.
- Fitness enthusiasts who want to organize general workout ideas.
- Educators demonstrating practical AI recommendation systems.



5. REQUIREMENTS SPECIFICATION

Hardware Requirements

Component	Minimum Requirement
Processor	Dual-core processor or better
RAM	4 GB minimum; 8 GB recommended
Storage	At least 2 GB free space for development
Display	Standard HD display
Network	Stable Internet connection for AI/API requests

Software Requirements

Software	Purpose
Operating System	Windows / Linux / macOS
Frontend	HTML, CSS, JavaScript or selected framework
Backend	Python / Node.js or equivalent
AI Service	Gemini API / supported Gemini model
IDE	Visual Studio Code or equivalent
Browser	Chrome, Edge, Firefox or equivalent

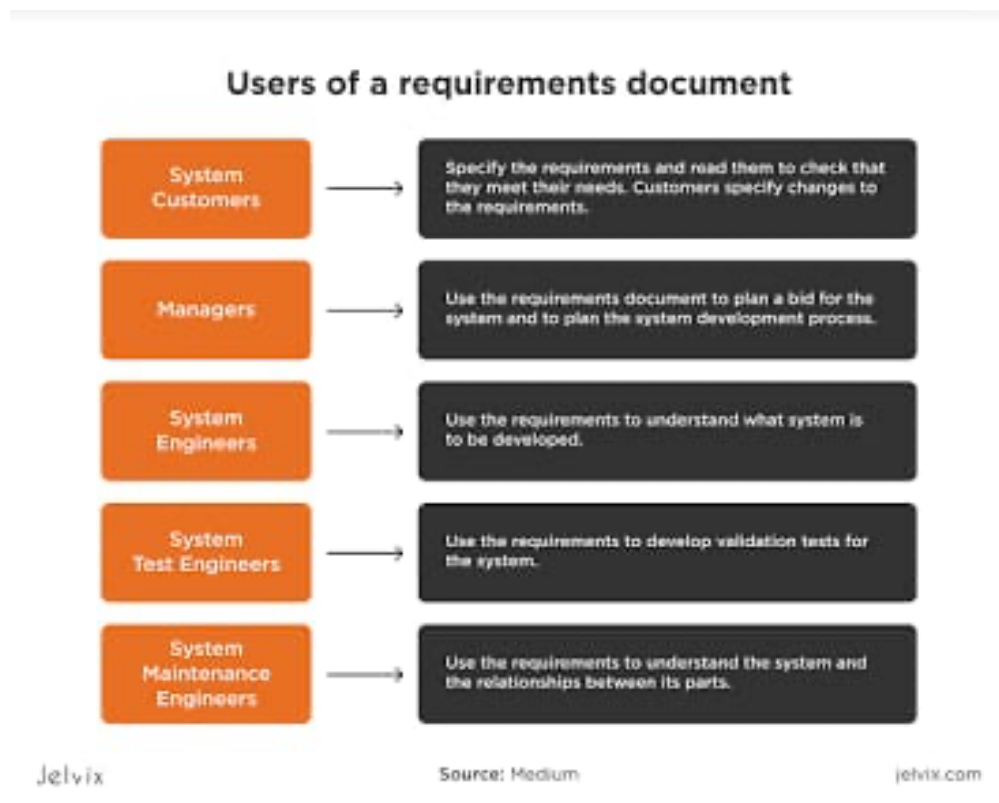
Functional Requirements

- User can enter a fitness goal.
- User can provide preferred duration and frequency.
- User can select experience level and exercise preferences.
- System validates required inputs.
- System builds a structured Gemini prompt.
- System displays the generated plan in a readable format.
- User can request modifications or regeneration.
- User can save or export the plan if implemented.

Non-Functional Requirements

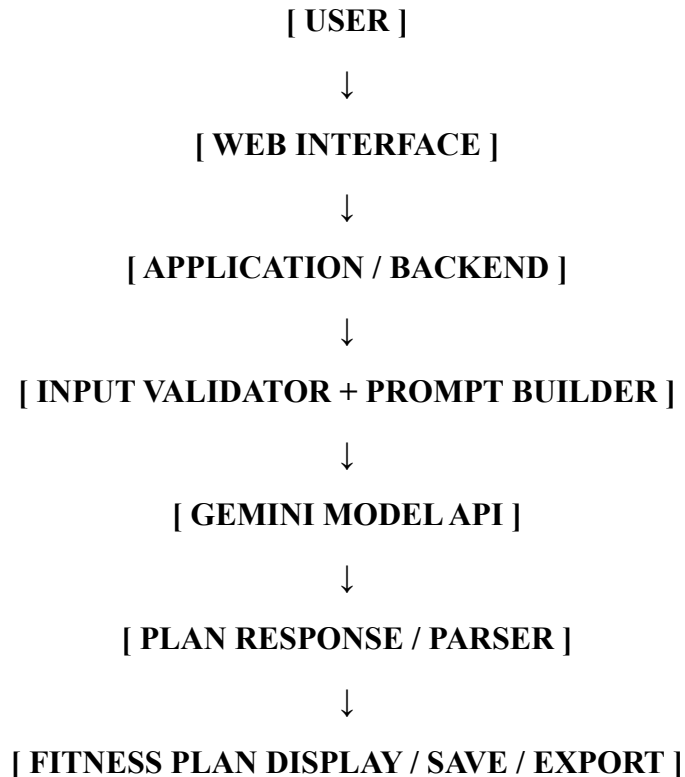
- Usability
- Performance

- Reliability
- Security
- Maintainability
- Scalability
- Responsible AI use



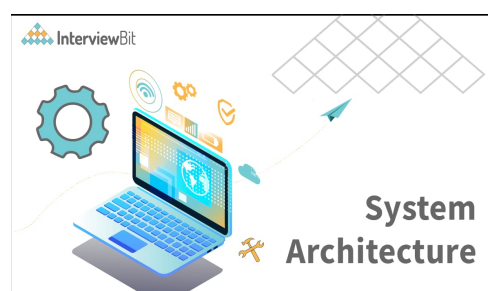
6. SYSTEM ARCHITECTURE

FITBUDDY consists of a presentation layer, application layer, prompt and validation layer, Gemini integration layer, and optional storage layer. The user enters planning preferences through the web interface. The backend validates the information and constructs a structured prompt. The Gemini service generates the response, which is processed and displayed by the application.



Architecture Components

- User Interface: Collects fitness goals and preferences.
- Input Validator: Checks required fields and acceptable values.
- Prompt Builder: Converts inputs into a detailed AI instruction.
- Gemini API Connector: Sends requests and receives generated content.
- Response Parser: Organizes the response into sessions or weekly sections.
- Storage Layer: Optionally stores plans and revisions.



7. MODULE DESCRIPTION

Module 1 – User Interface

Provides forms for fitness goal, experience level, preferred activity, session duration, frequency, available equipment, and other planning preferences.

Module 2 – User Input Validator

Checks required fields, reasonable ranges, missing information, and input length before the AI request is created.

Module 3 – Prompt Manager

Builds a structured instruction for the Gemini model and clearly specifies the requested plan format.

Module 4 – Gemini Integration

Connects the backend to the Gemini service and handles requests, responses, API errors, and configurable parameters.

Module 5 – Plan Generator

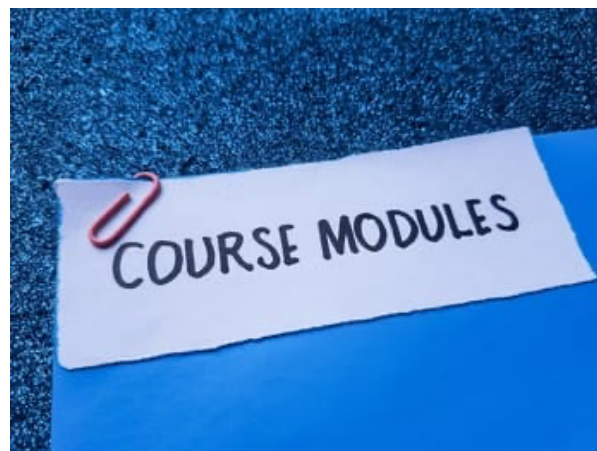
Transforms the generated response into a readable session-based or weekly plan.

Module 6 – Refine / Regenerate

Allows the user to request changes such as shorter sessions, different exercise preferences, or a revised weekly schedule.

Module 7 – Save / Export

Optionally stores generated plans or exports them as text/PDF or another project format.



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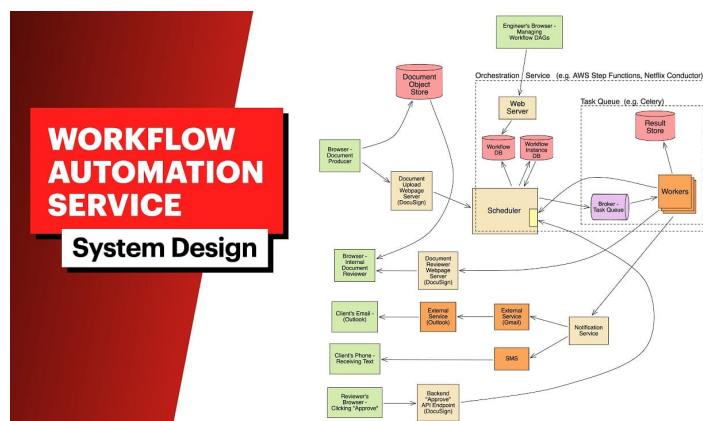
8. SYSTEM DESIGN AND WORKFLOW

Workflow

8. User opens FITBUDDY.
9. User enters a goal and planning preferences.
10. The system validates the information.
11. The prompt builder creates a structured Gemini request.
12. The request is sent to the configured Gemini model.
13. The generated response is received.
14. The response is validated and organized into a plan.
15. The user reviews the generated plan.
16. The user can request refinements.
17. The final plan can be saved or exported if the feature is enabled.

Example Input

“Create a beginner-friendly weekly general-fitness plan with short sessions, limited equipment, and clear warm-up and cool-down sections. Keep the output easy to understand.”



Expected Plan Structure

Section	Content
Goal	User's selected fitness objective
Weekly Schedule	Days and session structure
Warm-up	General preparation activities
Main Session	Suggested activities and organization
Rest / Recovery	Rest or lighter activity guidance
Cool-down	General cool-down suggestions
Notes	Safety and personalization reminders

9. DATA MODEL

A simple FITBUDDY prototype can operate without a permanent database. A complete application can store user projects, preferences, generated plans, and revision history in a database.

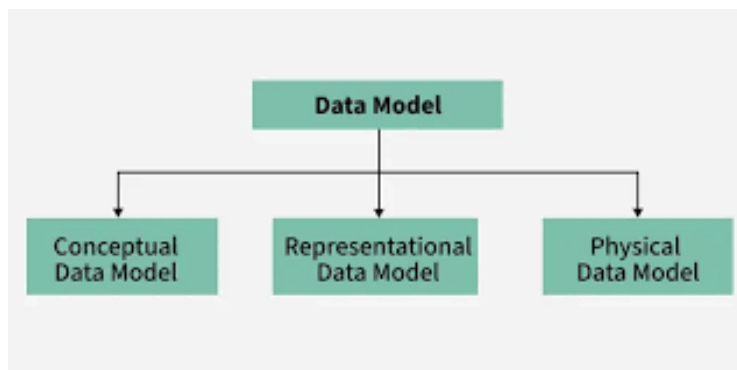
Entity	Important Fields
User	user_id, name, email, created_at
FitnessProfile	profile_id, user_id, goal, experience, preferences
Plan	plan_id, profile_id, prompt, generated_plan, version
Session	session_id, plan_id, day, activities, duration
Activity	activity_id, session_id, name, notes, duration
Revision	revision_id, plan_id, instruction, version, timestamp

Sample Plan Structure

Plan: { "goal": "General Fitness", "level": "Beginner", "days": 3, "session_duration": "30 minutes", "sessions": [...] }

Data Security Considerations

- API keys must not be exposed in client-side code.
- Collect only information required for the application's purpose.
- Protect saved user projects using appropriate access controls.
- Validate and sanitize user input.
- Do not present generated output as professional medical advice.



10. GEMINI MODEL INTEGRATION

Gemini models form the Generative AI component of FITBUDDY. The application uses structured prompts to convert user preferences into an organized fitness-plan response.

Prompt Engineering Strategy

- Define the model as an AI fitness-planning assistant.
- Include the user's stated goal, experience level, time, frequency, and preferences.
- Specify the desired output structure.
- Request clear session organization and readable language.
- Ask the model to include general safety reminders and encourage appropriate professional advice where relevant.
- Use refinement prompts when the user requests changes.

Illustrative Prompt Template

You are an AI fitness planning assistant. Based on the user's stated goal and preferences, create a general fitness-plan suggestion. Organize it by day or session and include warm-up, main activity, recovery/rest guidance, cool-down, and brief safety notes. Do not diagnose medical conditions or claim to replace a qualified professional. Follow the requested duration, experience level, and format.

Response Handling

The application should treat model output as generated external content. Responses should be validated before display, and unexpected or incomplete outputs should be handled gracefully. If a structured schema is used, the parser should verify required fields before rendering the plan.

```
from google import genai
```

```
client = genai.Client()
```

```
response = client.interactions.create(
```

```
    model="gemini-3.8-flash", input="Explain how AI works in a few words"
```

```
)
```

```
print(response.output_text)
```

```
...
```

<FollowUp>

If you'd like, let me know:

- * What **programming language or framework** you are using (e.g., Python, Spring AI, React)
- * Whether you need a **client-side or server-side** setup

I can provide the exact dependency configuration and code for your stack.

</FollowUp>

11. IMPLEMENTATION

Frontend Implementation

The frontend can provide a simple dashboard containing goal selection, experience level, session duration, weekly frequency, equipment preference, and an optional free-text prompt. A Generate Plan button sends the information to the backend.

Backend Implementation

The backend receives the request, validates inputs, builds the Gemini prompt, communicates with the configured Gemini API, processes the response, and returns the structured plan to the frontend.

Pseudo Code

START

Read fitness goal

Read experience level

Read duration and frequency

Read preferences/equipment

Validate input

Build structured Gemini prompt

Send request to Gemini model

Receive response

Validate response

Parse plan

Display sessions and notes

IF user requests changes

 Build refinement prompt

 Send updated request

 Display revised plan

END IF

Save/export if requested

END

User Interface Principles

- Use simple labels and clear input controls.
- Show a loading indicator during AI generation.
- Present each day/session in a separate section or card.
- Provide an easy refinement option.
- Display appropriate safety notices without making the interface confusing.

12. TESTING AND TEST CASES

Testing verifies that FITBUDDY correctly handles user inputs, generates structured plans, responds to refinement requests, and handles API or validation errors.

Test ID	Test Case	Expected Result	Status
TC01	Enter valid fitness goal	Plan is generated	Pass
TC02	Empty required field	Validation message displayed	Pass
TC03	Select beginner level	Output reflects beginner-oriented planning	Pass
TC04	Change session duration	Generated plan follows requested duration	Pass
TC05	Change weekly frequency	Schedule follows requested frequency	Pass
TC06	Refinement request	Plan is regenerated with requested change	Pass
TC07	API/network failure	Clear error message displayed	Pass
TC08	Unexpected model response	Application handles response safely	Pass
TC09	Save/export plan	Plan is saved/exported if enabled	Pass
TC10	Mobile browser	Interface remains usable	Pass

Testing Methods

- Unit testing for validation and formatting functions.
- Integration testing for frontend, backend, and Gemini API.
- System testing for complete input-to-plan workflow.
- Usability testing with representative users.
- Negative testing using missing, excessive, and unexpected inputs.

13. RESULTS, ADVANTAGES AND LIMITATIONS

Expected / Demonstrated Result

The FITBUDDY prototype demonstrates an end-to-end AI-assisted fitness planning workflow. User preferences are converted into a structured prompt, processed by a Gemini model, and displayed as an organized fitness-plan suggestion that can be refined interactively.

Advantages

- Provides a simple interface for generating structured fitness plans.
- Supports different goals and user preferences.
- Allows rapid plan refinement through natural-language prompts.
- Reduces manual effort in organizing sessions.
- Demonstrates a practical Generative AI recommendation use case.
- Can be extended with progress tracking and multimedia features.

Limitations

- AI-generated plans may contain inappropriate or unsuitable suggestions and require user review.
- The system cannot reliably assess physical condition without professional evaluation.
- Individual response to exercise varies.
- API availability, quotas, and costs can affect operation.
- The project is not intended to diagnose, treat, or manage medical conditions.
- Generated content may require verification and personalization by a qualified professional for users with specific concerns.

Responsible AI Considerations

The application should clearly communicate that AI output is general planning assistance. It should avoid diagnosing injuries or diseases, avoid unsupported medical claims, and encourage professional guidance where a user's situation requires it.

14. CONCLUSION AND FUTURE ENHANCEMENT

Conclusion

FITBUDDY – AI FITNESS PLAN GENERATOR Using Gemini Models is a Generative AI based project that demonstrates how an AI model can assist users in organizing general fitness activities according to stated goals and preferences. The system combines web development, prompt engineering, recommendation concepts, and Gemini API integration.

The project provides an interactive workflow in which users can generate, review, and refine a structured plan. It also demonstrates important software-development considerations such as input validation, response parsing, error handling, security, usability, and responsible AI communication.

FITBUDDY should be considered an educational and general planning application rather than a replacement for qualified medical or fitness professionals. This distinction is important when designing AI applications in health and wellness-related domains.

Future Enhancements

18. Add exercise demonstration images or videos from appropriate sources.
19. Add progress tracking and weekly activity history.
20. Add reminders and calendar integration.
21. Add multilingual support, including Tamil and English.
22. Add wearable-device integration where technically and ethically appropriate.
23. Add user feedback to improve plan refinement.
24. Add secure user accounts and cloud project storage.
25. Add nutrition-planning features with appropriate safety boundaries.
26. Add accessibility improvements for different user needs.

References

- Google Gemini / Google AI official documentation – Gemini model and API concepts.
- Natural Language Processing and Generative AI academic references.
- HTML, CSS and JavaScript documentation – web application development.
- Python / Node.js documentation – backend development, depending on implementation.
- General fitness and exercise guidance from qualified professional or public-health resources, where incorporated into the final application.